

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a machine-checked synthesis of established physical
8 results into a single conditional route from the finiteness of a local-
9 ized system’s phase space to a unified cognitive fixed point. No new
10 fundamental physics is proposed: Kibble’s classification of topolog-
11 ical defects, Landauer’s erasure bound, the bound of Still and col-
12 leagues relating dissipation to nonpredictive information, the thresh-
13 old $K_c = 2D$ for the noisy mean-field Kuramoto model, and Banach’s
14 fixed-point theorem are all imported. What is new is that they are
15 stated in one formal language, their mutual consistency mechanically
16 checked, and the points at which the route relies on stipulation rather

17 than derivation located exactly. One conditional theorem in Lean 4
18 composes eight named hypotheses to its endpoint; the unproved steps
19 are its arguments rather than sentences about it, and they are jointly
20 satisfiable rather than jointly empty. Two links are marked rather
21 than argued around: the identification of the posited continuum ker-
22 nel with the endogenous electromagnetic field of cortex is an empirical
23 commitment with four stated failure conditions, and the identification
24 of a global section of the probability sheaf with the unity of experience
25 is a stipulation. The framework is falsifiable in one place that is its
26 own. Recovery from sudden awakening should follow a delayed sig-
27 moid with a square-root cusp rather than an exponential, because the
28 slow variable gating recovery is astrocytic volume rather than cortical
29 phase realignment, and throughout that recovery the concentration
30 and the coherence of the phase distribution must collapse onto the
31 parameter-free curve $r = I_1(a)/I_0(a)$.

32 1 Introduction

33 The pursuit of a physical theory of consciousness typically involves search-
34 ing for the biological correlates of mind or treating subjective experience
35 as an abstract computational algorithm. This paper asks a more funda-
36 mental question: if we assume only basic physical primitives—continuous
37 symmetries, geometry, and thermodynamics—is the emergence of a unified,
38 perceiving entity a natural physical attractor?

39 We propose a framework in which the structural properties of conscious-
 40 ness are organized as candidate physical attractor states, running from con-
 41 tinuous symmetries and boundaries, through dissipation and physical reso-
 42 nance, to the macroscopic electromagnetic unity proposed by field theories
 43 of mind. This is not a derivation from basic principles: the formal result is
 44 a conditional composition schema whose eight unproved arrows are explicit
 45 assumptions. To avoid critiques of “math theatre,” we delineate between lit-
 46 eral physical mechanisms (thermodynamic erasure via Landauer’s principle,
 47 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 48 using Lean 4 formalization as our mathematical anchor. We adopt a perspec-
 49 tive aligned with Russellian monism: objective measurements of macroscopic
 50 electromagnetic fields constitute the third-person “outside,” while the sys-
 51 tem’s internal physical deformation to reach structural resonance constitutes
 52 the first-person intrinsic experience.

53 **What is new here, and what is not**

54 A framework paper about consciousness invites a particular suspicion, and it
 55 is better answered at the start than at the end. **No new physics is derived**
 56 **here.** Every physical result the chain invokes is established, and most of
 57 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 58 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 59 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 60 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative

61 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
62 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
63 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
64 Bell and Crooks relating dissipation to nonpredictive information (Still et
65 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
66 dissipated work with the relative entropy between forward and time-reversed
67 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
68 1922). A physicist reading this paper will recognize all of it. Conceding the
69 point costs nothing.

70 What is new is of a different kind, and it is a smaller claim. These results
71 are stated in one formal language, their mutual consistency mechanically
72 checked, and the places where the chain relies on stipulation rather than
73 derivation located exactly. The development declares **no axioms: #print**
74 **axioms** reports only **propext**, **Classical.choice** and **Quot.sound**. Ev-
75 ery physical postulate is a field of a class each model discharges for itself,
76 and every class is inhabited by an explicit witness, so no theorem is vac-
77 uous. The links form a single conditional composition theorem — **chain**
78 in **extttChain.lean** — with **eight named hypotheses**: four formalization
79 gaps, two modelling assumptions, two physical commitments. Its conclu-
80 sion **extttUnifiedSelf** says the sheaf-glued section is the unique fixed point
81 of self-prediction, rather than **Unity** and **Self** being unrelated. The theo-
82 rem **chain_hypotheses_jointly_satisfiable** discharges all eight on one
83 toy substrate, ruling out inconsistency without evidence that cortex realizes

84 the composition.

85 The schema exposes two missing arrows: the capacity bound implies the
86 heat cost directly, skipping the boundary, and neither of the two theorem-
87 arrows is one the framework’s usual narration draws.

88 The framework is not thereby unfalsifiable, and one of its predictions is
89 its own rather than a rival’s. Section 12 argues that recovery from sudden
90 awakening should follow a *delayed sigmoid* rather than an exponential, be-
91 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
92 clearance, on a timescale of seconds to minutes, while the cortical rhythms
93 whose realignment is usually taken to constitute the recovery move in mil-
94 liseconds. That is a claim about functional form; it is fittable against data
95 of a kind already collected; and a purely electrical account of the same re-
96 covery predicts a different form. Because a delayed sigmoid is what any slow
97 saturating gate would give, that section also states the sharper claim the
98 same mathematics supports: at every instant of the recovery the concentra-
99 tion and the coherence of the phase distribution must lie on the fixed curve
100 $r = I_1(a)/I_0(a)$, the same curve for every subject, with no free parameters
101 and with the threshold read off the ratio a/r rather than fitted.

102 **2 The chain, and the status of each of its links**

103 The machine-checked dependency accounting is carried out in Lean 4, build-
104 ing on recent efforts to formalize consciousness and brain semantics such as

105 the ‘OnlyOne’ system (Scherf, 2026) and sheaf-theoretic models of neural
 106 manifolds (Inoué, 2026). Figure 1 shows the schema with the status of every
 107 node *and of every arrow* marked, and Table 1 states in seven rows what
 108 each link assumes. The complete claim-by-claim map to Lean identifiers is
 109 Table S1 in the supplement; the main text does not depend on reading it.

110 One result about the formalization itself is worth stating here rather than
 111 in the supplement, because it is what makes the accounting mean anything.
 112 An axiom that is inconsistent silently makes every theorem downstream of
 113 it vacuous while leaving the build green, and the failure has one shape: *an*
 114 *axiom that constrains a symbol it does not itself bind*. Such a declaration
 115 is not a postulate about one system but a constraint on every model of the
 116 class the symbol belongs to, including models built to violate it, and a class
 117 that has such a model is thereby inconsistent. Three tempting global axiom
 118 shapes — pointwise vacuum minimization with the energy functional as an
 119 implicit argument, a Landauer heat identity equating a free class field with a
 120 bath-entropy difference for every instance, and an entropy-production bound
 121 asserted against all pairs of distributions — each prove **False**, from that one
 122 root cause. The rule that prevents it is the rule we follow throughout: *a*
 123 *physical postulate that mentions a class field must be a field of that class,*
 124 *never a standalone axiom quantified over all instances*. A postulate is then
 125 an obligation each model discharges, not a global claim about every model.
 126 The supplement gives the three refutations and the corresponding witness
 127 constructions, which are what keep the classes inhabited and their theorems

128 non-vacuous.

129 3 Symmetries as setting, and the capacity of 130 a finite phase space

131 Our starting premise relies on fundamental geometry. Continuous spacetime
132 symmetries act upon field states, and the invariance of the system's total
133 energy under that action is recorded by the class `SymmetryInvariantAction`
134 (`Phase1Primitives.lean`). Physically, it is such an invariance that Noether's
135 theorem converts into the conservation of energy and momentum. The scope
136 of the *formal* claim is narrower and we state it plainly: the invariance is a
137 class field, an assumption a model must discharge, and no conserved quantity
138 is ever constructed from it. There is no Noether theorem in our Lean, and
139 nothing downstream depends on one. The class is inhabited by a double-
140 well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime with the $\mathbb{Z}_2$ action
141 $v \mapsto \pm v$, whose vacuum manifold $\{-1, 1\}$ is proved degenerate with the two
142 minima exchanged by the symmetry. That is symmetry breaking exhibited
143 rather than assumed; it is still not a conserved quantity. Symmetries and the
144 metric are therefore *setting* rather than content, and Figure 1 marks them
145 as such.

146 The other half of the premise is not setting. Because phase space is finite,
147 any localized physical system possesses a mathematical limit on the informa-
148 tion it can embody at any moment, and three theorems in `Phase1PhaseSpaceCapacity.lean`

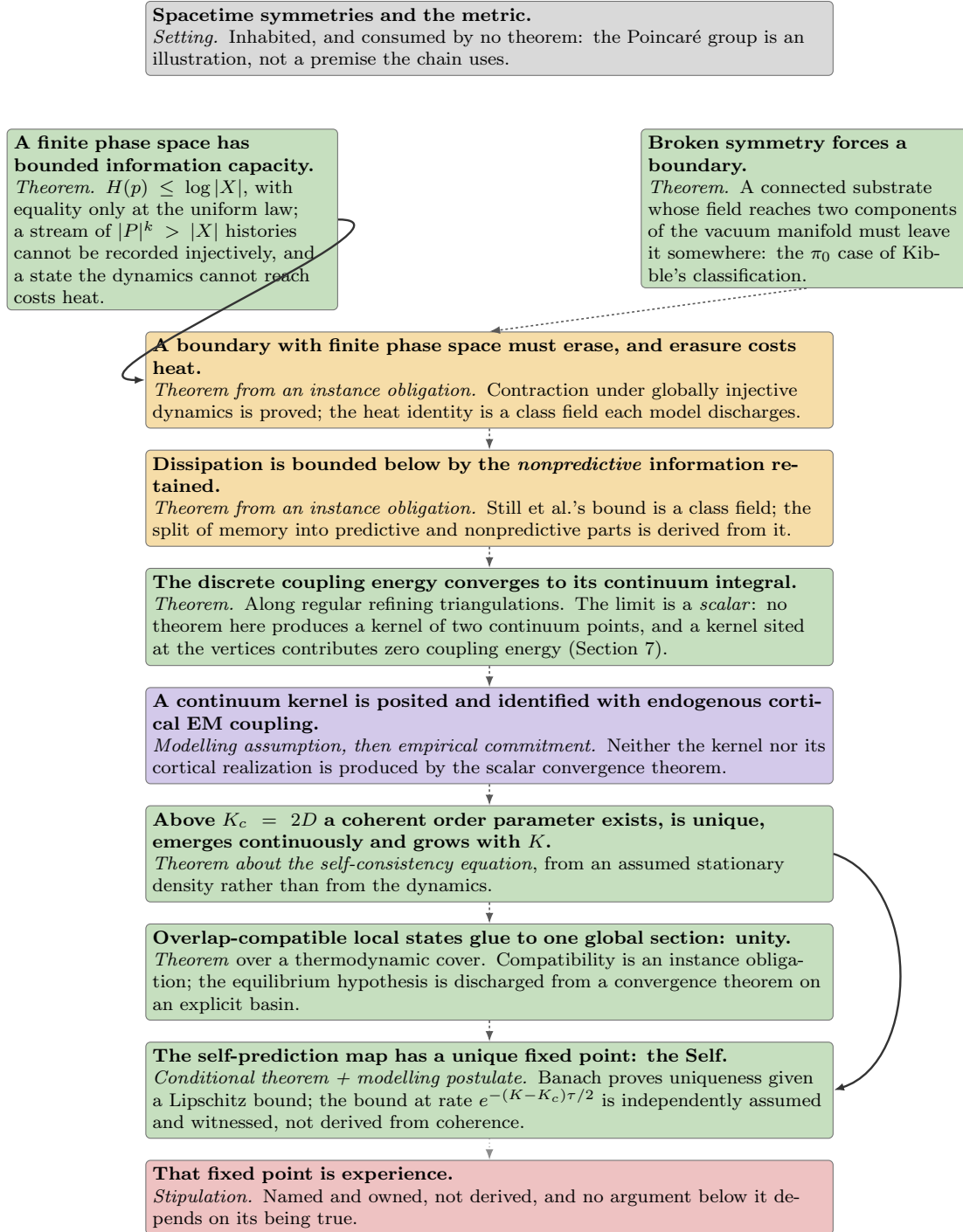


Figure 1: The conditional composition schema, with the status of every node and the distinction between the two kinds of arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of `chain` (`extttChain.lean`): an implication between the two node statements which the development does not prove and which is therefore an argument the theorem takes. A **dotted** ar

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasure costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and overlap-compatible patches glue into a unique global section	Theorem 9	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.
The coherent order parameter carries a unique reflexive fixed point:	Conditional theorem + modelling	The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate

149 carry it, whose only hypothesis about the system is that its state space is
150 finite.

151 The first is the capacity bound. A probability distribution p on a fi-
152 nite state space X satisfies $H(p) \leq \log |X|$ (`shannon_entropy_le_log_card`),
153 with equality if and only if p is uniform (`shannon_entropy_eq_log_card_`
154 `iff`). The “only if” half is the part worth stating: $\log |X|$ is the exact capac-
155 ity rather than a convenient over-estimate. The proof is Gibbs’ inequality
156 against the uniform law, with the $p(x) = 0$ case carrying the slack that makes
157 the equality condition sharp.

158 The second says the limit binds. Let the boundary absorb a stream of
159 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
160 possible histories, and once $|X| < |P|^k$ the map from histories to final states
161 cannot be injective (`absorb_not_injective`), so two distinct histories leave
162 the system in the same state (`source_entropy_exceeds_capacity`). The
163 update map is arbitrary and may be reversible at every individual step; the
164 collision is forced by counting alone, which is the sense in which the premise,
165 and not the model of the dynamics, does the work.

166 The third is what the thermodynamic step consumes. On a finite state
167 space, a map that fails to reach every state is not injective (`is_erasure_of_`
168 `not_surjective`) — which is exactly the erasure condition Landauer’s bound
169 requires — so any update of a finite phase space leaving some state unreach-
170 able dissipates strictly positive heat (`finite_phase_space_dissipates`). On
171 an infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses

172 0 and is injective. That failure is recorded beside the witnesses, so the finite-
173 ness hypothesis is fenced rather than assumed to be load-bearing. None of
174 this derives the finiteness of phase space; finiteness is assumed, as in any
175 statistical-mechanical treatment. What is removed is the gap between as-
176 suming the premise and using it.

177 4 Symmetry breaking and the inevitability of 178 boundaries

179 A perfectly symmetric universe is the uniform vacuum; no distinct entities
180 exist. For any dynamic process to occur, the continuous symmetries of the
181 background field must undergo spontaneous symmetry breaking, and topol-
182 ogy then dictates the creation of defects such as domain walls. This neces-
183 sitates a physical boundary—a localized topological distinction between an
184 “inside” and an “outside.” Such a defect is necessary to define a separate
185 entity but not sufficient for agency: the boundary is relational, defined by
186 the twisted geometry of the field surrounding it, and must be coupled with
187 non-equilibrium thermodynamics to exhibit active behaviour.

188 That first claim is about topology, and it needs a different argument from
189 the energy claim beside it. The energy half — an energy-minimizing field sits
190 in the vacuum manifold almost everywhere (`pointwise_vacuum_of_global_`
191 `min`) — together with the observation that a defect *obstructs* extension over
192 a contractible interior, produces no defect: both are statements about a

193 field already given. What produces one is `exists_notMem_of_connected-`
 194 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 195 ues in two *different connected components* of the vacuum manifold M , then
 196 somewhere the field is not in M at all. That excluded region is the wall.
 197 This is the π_0 case of Kibble’s classification (Kibble, 1976), and the proof
 198 is elementary in exactly the way it is elementary in the physics literature:
 199 the continuous image of a connected space is connected, and a connected
 200 subset of M lies inside one component of it. No hypothesis mentions energy,
 201 a potential, or a symmetry.

202 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 203 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are available in Mathlib’s
 204 homotopy library and not taken, because the argument this paper runs uses
 205 domain walls and nothing else. The witness is the double well, where the vac-
 206 uum manifold is *computed* to be exactly $\{-1, 1\}$ rather than merely known to
 207 contain both minima, and its disconnection is proved in the form the theorem
 208 consumes. `wellV_domain_wall` then quantifies over *every* continuous field on
 209 *any* connected substrate reaching both minima, with no formula for the field
 210 in the statement; that the class is non-empty is checked separately, since a
 211 theorem quantified over all fields is worthless if none satisfies its hypotheses.

212 5 The thermodynamic constraint

213 This topological boundary is subjected to gauge perturbations from the “out-
214 side” environment. Because the internal phase space is finite, it cannot ab-
215 sorb an unbounded sequence of incoming perturbations and must overwrite
216 old internal states. By Landauer’s principle (Landauer, 1961), the erasure of
217 physical information irreversibly decreases internal entropy, mandating that
218 the system exhaust heat into the environment.

219 Norton and Shenker have argued that *logical* erasure (many-to-one on
220 abstract computational states) need not dissipate heat if implemented by an
221 injective physical process (Norton, 2005; Shenker, 2000). Our formalization
222 addresses this by defining erasure as a non-injective mapping on the *physical*
223 state space and formalizing the global system–bath evolution as strictly in-
224 jective (Liouville). Heat dissipation is then a consequence of local state-space
225 contraction under globally reversible dynamics. The finiteness premise enters
226 exactly here: on a finite phase space, a state becoming unreachable is the
227 same as non-injectivity, and `finite_phase_space_dissipates` is the result.
228 The Norton–Shenker distinction is preserved rather than blurred: an update
229 that keeps the incoming perturbation on record is injective and dissipates
230 nothing, while one that overwrites it is not. Converting the combinatorial
231 statement to heat requires the class field `StatisticalMechanics.heat_eq`,
232 discharged by an explicit one-bit erasure system.

233 6 Prediction as the thermodynamic cost of 234 memory

235 A dissipative structure is driven by its environment. Write X_t for the state
236 of the structure and S_t for the environmental signal driving it, and let the
237 signal evolve by a dynamics of its own on which the structure does not act.
238 Two quantities are then defined at each step: the *memory* $I(X_t; S_t)$, and the
239 *predictive information* $I(X_t; S_{t+1})$, which is what that same state says about
240 the signal's next value.

241 **What the entropy-production bound does not give.** There is a nat-
242 ural route from entropy production to predictive processing, and it does not
243 work. Modelling the environment as a stochastic driving force $P(\text{ext})$ and
244 the internal transitions as $Q(\text{int})$, entropy production is bounded below by
245 $D_{KL}(P||Q)/\Delta t$ — this is Gibbs' inequality, carried in Lean as an instance
246 obligation. Rearranged it reads $D_{KL} \leq \Delta t \sigma$, so a system descending on σ
247 lowers a *bound* on the divergence, and it is tempting to conclude that the
248 internal transition probabilities come to mirror the environment's statistical
249 symmetries. They do not, because a driven system does not relax to $\sigma = 0$:
250 it relaxes to a non-equilibrium steady state at strictly positive entropy pro-
251 duction, and the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number.
252 No limit may be read off it.

253 **Where predictive processing comes from.** Still, Sivak, Bell and Crooks
 254 (Still et al., 2012) prove that for a system driven by an environmental signal
 255 the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

256 The bracket is the *nonpredictive* information: what the structure retains
 257 about the signal’s present that says nothing about its future. Memory
 258 that fails to predict is not merely useless to the structure; it is thermo-
 259 dynamic waste, quantified in energy. The bound is carried as the class field
 260 `PredictiveDissipation.still_bound`, for the same reason as every other
 261 physical postulate here: its `axiom` form, quantified over all joint laws with
 262 the dissipated work fixed, is refutable, and the development refutes it.

263 What is *derived* is everything downstream. First, the bracket is non-
 264 negative: $X_t \rightarrow S_t \rightarrow S_{t+1}$ is a Markov chain, so the data processing inequal-
 265 ity gives $I(X_t; S_{t+1}) \leq I(X_t; S_t)$ (`predictiveInfo_le_mutualInfo`). Second,
 266 dissipation is bounded below by a non-negative quantity, so $W_{\text{diss}} \geq 0$ follows
 267 rather than being assumed (`dissipatedWork_nonneg`). Third, the inference
 268 runs in the valid direction: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$, so any bound
 269 on dissipation is immediately a bound on wasted memory (`nonpredictive_`
 270 `le_dissipation`). In the idealised quasi-static case every retained bit is
 271 predictive.

272 The no-back-action hypothesis is load-bearing and we exhibit what breaks

273 without it. In the formalization it is not an assumption stated beside the
 274 theorem but the shape of the evolution itself: the signal's step is the parallel
 275 kernel id $\parallel \kappa$, so no channel from X_t to S_{t+1} exists. A Markov kernel on the
 276 *pair* which copies the system's state into the signal carries an independent
 277 two-bit law to a perfectly correlated one, raising mutual information from
 278 zero to a positive value: an agent that steers its environment can know
 279 more about that environment's future than about its present, and Still's
 280 bound is not about such an agent. Both regimes are exhibited over one
 281 correlated two-bit joint law. With the signal frozen, every remembered bit
 282 is predictive and the instance may dissipate nothing, with the memory itself
 283 proved strictly positive; with the signal redrawn from its stationary law at
 284 each step, the entire memory is nonpredictive and the instance is *forced* to
 285 dissipate $k_B T$ times all of it. Kawai, Parrondo and Van den Broeck (Kawai
 286 et al., 2007) supply the corresponding equality — average dissipated work is
 287 exactly $k_B T$ times the relative entropy between forward and time-reversed
 288 phase-space densities — carried here as a predicate on an instance, which
 289 replaces the abstract dissipation field with a quantity an experiment can
 290 estimate (`nonpredictive_le_arrow`).

291 **What this does not establish.** It does not deliver the Free Energy Prin-
 292 ciple. Still's bound says that prediction is thermodynamically favoured; it
 293 says nothing about hierarchical generative models, variational free energy,
 294 or Bayesian inference. The Free Energy Principle (Friston, 2010; Friston &

295 Kiebel, 2009; Hohwy, 2013; Clark, 2016; Spisák & Friston, 2026) and the
 296 General Resonance Theory of Consciousness (Hunt & Schooler, 2019) are
 297 consonant with what is proved here, which is a weaker relation than deriva-
 298 tion and the only one we assert. Nor does it solve the symbol grounding
 299 problem (Harnad, 1990): maximal predictive information is compatible with
 300 the structure storing a lossy or unrecognisable function of the signal, and a
 301 state that predicts is not thereby a state that means. The narrower claim is
 302 the one worth making: whatever internal states a physical structure holds,
 303 those that do not help predict its environment are paid for in heat, so the
 304 pressure toward states indexed to the environment is energetic rather than
 305 semantic. We call the phenomenon *structural resonance*, and there is no
 306 teleological drive to “survive” in it: predicting the environment is simply the
 307 cheaper of the two ways to hold a memory.

308 A separate gap lies between the stationary threshold theorems of Deriva-
 309 tion 10 and any statement about trajectories: the threshold $K_c = 2D$ is
 310 proved for a stationary density, not for the finite-oscillator system, and the
 311 dynamical mean-field limit — propagation of chaos for the Kuramoto sys-
 312 tem — is the missing link between them (Lacker, 2016; Sznitman, 1991). No
 313 theorem in this development closes it.

314 7 Macroscopic scaling: a theorem and a com- 315 mitment

316 Three steps of different kinds meet here and are easy to run together: con-
317 vergence of a scalar energy, which is a theorem; introduction of a continuum
318 coupling kernel, which is a modelling assumption; and identification of that
319 kernel with the brain’s endogenous electromagnetic field, which is an empir-
320 ical claim about cortex. Neither later step inherits the licence of the first.
321 Kept apart, the empirical claim is the framework’s central commitment and
322 the place where it is most easily wrong — which is where a claim of this kind
323 belongs.

324 **The formal half.** `Phase2_MeshConvergence.lean` proves that the discrete
325 coupling energy converges to a continuous integral along regular refining
326 triangulations (`mesh_refinement_convergence`). Because the summand is
327 symmetric, the double sum is the midpoint rule over unordered edges, so the
328 discrete energy is a quadrature of the continuous one. What this does not
329 establish: it is a statement about an energy functional, not about dynamics
330 — no trajectory convergence and no renormalization-group flow. The con-
331 tinuous object, `ContinuousNeuralField`, is *posited* with a coupling kernel
332 K , not produced by the theorem.

333 **The empirical step.** The formal half supplies scalar energy convergence
334 and no kernel. The field sections therefore posit a symmetric continuum

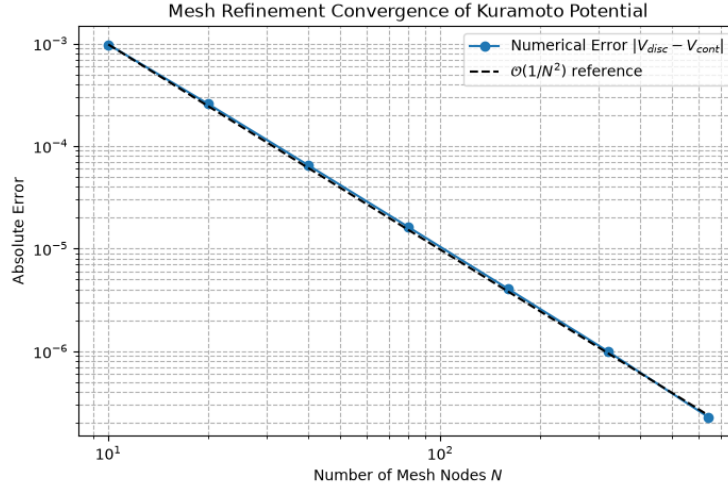


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones. Parameters and scripts are in the supplement.

335 kernel as a modelling input. Which physical variable in cortex realizes that
 336 independently supplied kernel is a further empirical question, and we state
 337 our answer as a commitment:

338 *The model identifies its posited continuum kernel with the endoge-*
 339 *nous electromagnetic field of mammalian cortex.*

340 This is a claim about biology, not a consequence of anything above it.
 341 It is what makes the framework falsifiable and it is what we expect to be

342 attacked.

343 **What supports it.** Neurons do not only signal through synapses: trans-
344 membrane currents generate extracellular fields that shift membrane poten-
345 tials of neighbouring cells, a mechanism known as ephaptic coupling. En-
346 dogenous fields of 1–5 mV/mm entrain spike timing (Fröhlich & McCormick,
347 2010; Anastassiou & Koch, 2015), and stable electric fields carry information
348 that survives representational drift (Pinotsis & Miller, 2022). The field is
349 spatially continuous and superposes contributions from large populations —
350 a medium in which spatially separated activity is combined without further
351 mechanism. This is the integrative role the CEMI field theory assigns it
352 (McFadden, 2002, 2020; Pockett, 2012), physically supported by astrocytic
353 syncytia joined by gap junctions (Verkhatsky & Nedergaard, 2018).

354 **What the evidence supports, and what it does not.** The endogenous
355 field’s effect on any individual neuron is modulatory — it biases when a cell
356 fires; it does not decide whether it fires (Anastassiou & Koch, 2015). We
357 do not claim it is the *primary* dissipative structure. The claim is that it is
358 modulatory in effect and integrative in spatial reach: weak per synapse, and
359 the only variable defined everywhere at once. What the framework needs is
360 the reach, not the strength.

361 **Why reach is the property that matters.** A theory of consciousness
362 needs an account of why many local processes make one experience, and most

supply it by stipulation or by a measure. The account here is structural, and it is the framework’s strongest argument. The field is the only variable in the system that is *defined everywhere at once*. Unity is not asserted of it; unity is what a field is. That is what makes the sheaf-theoretic step of Section 8 available at all — local states glue to one global section because there is one object over which gluing can happen — so the answer to the combination problem is a piece of geometry rather than a postulate. Compare “integration” in Integrated Information Theory, which is a number, or “broadcast” in Global Workspace Theory, which is a metaphor about a workspace. It is also what the empirical identification is carrying: if the posited continuum coupling is realized by something that is *not* defined everywhere at once — by synaptic connectivity, say — then this argument is unavailable, and the framework loses the property that distinguishes it from its rivals rather than merely losing a mechanism.

Its critics. The longest-standing objection is that the fields are too weak to matter, pressed from within the field-theoretic camp by Pockett herself (Pockett, 2012, 2002). Vöröslakos and colleagues found that applied current is shunted through scalp, delivering fields at or below 1 V/m in cortex (Vöröslakos et al., 2018). That is a bound on *applied* rather than endogenous fields, but it constrains how much causal work a field of that magnitude can do, and is the reason we do not claim the field drives spiking.

384 **What would falsify it.** We state the failure conditions rather than leav-
385 ing them to a referee. **Epiphenomenality:** if the endogenous field can
386 be cancelled or shunted without altering transmembrane currents and nei-
387 ther behaviour nor report changes, the identification fails. **Magnitude gap:**
388 if the spatial integration required demands field effects an order of magni-
389 tude above the 1–5 mV/mm measured in cortex, the commitment fails on
390 arithmetic. **Timescale mismatch:** if recovery from sleep inertia tracks an
391 exponential rather than the delayed sigmoid the slow variable predicts, the
392 framework loses its own prediction. **Dissociation:** if large-scale field coher-
393 ence is preserved where unity is lost — or absent where it is retained — the
394 identification has no physical carrier.

395 **Can a modulatory field cross the threshold?** The objection that 1–5
396 mV/mm is too weak to meet $K > K_c$ assumes that the per-neuron effect of
397 the field is what enters the mean-field coupling constant. It enters through
398 population arithmetic. A field of 3.0 mV/mm shifts spike timing by ~ 1.2
399 ms (Anastassiou et al., 2011; Anastassiou & Koch, 2015); at 40 Hz, that is
400 a per-neuron phase advance of $\Delta t \cdot f \approx 0.007$ per cycle. The LFP decay
401 length $\lambda \sim 0.2$ mm (Lindén et al., 2011) encloses $N \approx 1676$ neurons at
402 a cortical density of $5e + 04 \text{ mm}^{-3}$ (Herculano-Houzel, 2009). Integrating:
403 $K \sim N \Delta t f \approx 80$. Against a Lorentzian frequency-spread floor $D \sim 1.5$
404 rad/s (Strogatz & Mirollo, 1991; Buzsáki & Draguhn, 2004), the ratio $K/D \sim$
405 53.6 clears the threshold $K_c/D = 2.0$ by a wide margin. The estimate is

406 parametric in λ , which is empirically known and spans 0.1–0.3 mm (Lindén
 407 et al., 2011); across the full range and with field amplitudes at the measured
 408 bounds, the ratio stays above 30.2. Sweeping across canonical EEG bands
 409 with band-typical amplitudes (gamma: 0.5 mV/mm; beta: 1.0; alpha: 1.5;
 410 theta: 3.0; delta: 2.0), the same arithmetic yields K/D ranging from 8.9
 411 down to 1.8 — delta is the one band that sits below K_c , consistent with
 412 the loss of consciousness during slow-wave sleep. This is not a measurement,
 413 and it is the one commitment of the chain that is simplest to falsify on
 414 arithmetic alone: if the coupling parameters sit at the conservative end of
 415 each published range, K/D drops to 5.7 — still above threshold, but requiring
 416 the combination to survive independent measurement rather than being ruled
 417 out by the same arithmetic.

418 **The regime the theorems cover.** The dissipative account — well-posedness,
 419 the Lyapunov identity, monotone descent, and that the motion stops along
 420 every trajectory — holds with no assumption on the sign of the coupling.
 421 *Where* it stops does not, and the claims about the destination all live on
 422 the positive side. The memory capacity of a frustrated cortical network is
 423 therefore not supported by any theorem here, and we do not assert it as part
 424 of the chain. The natural repair — letting local states agree on overlaps up
 425 to a phase and asking whether the offset cocycle is a coboundary — fails
 426 because locked, twisted and splay configurations all assign a phase to every
 427 patch, so their offsets are automatically a coboundary.

428 Primate LFP phase maps contain phase singularities with integer winding
 429 number, and human cortical spiral centres classify cognitive tasks (Townsend
 430 et al., 2015; Xu et al., 2023). `Phase5.PhaseLifts` formalizes the measured
 431 loop: transitions from one global real lift telescope to zero, while a three-
 432 patch loop with winding one admits no global lift. The supplement gives the
 433 measurement protocol and preregistered test this supports — a falsifiable
 434 content test outside the conditional chain.

435 **8 Global sections and the unity of conscious-** 436 **ness**

437 The system is now characterized by a continuous macroscopic electrodynamic
 438 field generated by many coupled micro-components, forced to resonate with
 439 the environment. From here on the coupling regime is the unfrustrated one,
 440 which is what every theorem below requires. The Kuramoto model (Ku-
 441 ramoto, 1984) guarantees that past a critical coupling threshold these dis-
 442 tributed local fields undergo a phase transition into unified synchronization
 443 (Bauer et al., 2022), and `Phase4.KuramotoDynamics.lean` confirms that the
 444 phase-locked state minimizes the Lyapunov potential — in the precise sense
 445 that the functional minimized is the coupling potential read in the frame ro-
 446 tating with the natural frequency, which `Phase4.RotatingFrame.lean` shows
 447 is the honest reduced description of a system of identical oscillators. With
 448 any natural frequency non-zero the full potential is unbounded below, so it

449 has no minimum for phase-locking to attain; the reduction is exact only for
 450 identical frequencies, and a genuine spread leaves residual detunings whose
 451 threshold is nowhere in our development.

452 The phase transition supplies synchronized phases. A separate compat-
 453 ibility hypothesis says the corresponding local states agree on every over-
 454 lap; given that, the sheaf condition glues them into a singular *global section*
 455 (`Phase5_GlobalSection.lean`). The field acts as a single, indivisible topo-
 456 logical space, providing the *unity of consciousness*—a necessary condition
 457 for integrated experience. `ThermodynamicCover` demands that the cover’s
 458 phase field minimize the reduced Kuramoto potential, and `potential_min_`
 459 `iff_phase_locked` characterizes those minimizers exactly, in both directions.
 460 Every instance is therefore already phase-locked, and the class’s single con-
 461 dition on its local data — that patches at a common phase agree where
 462 they overlap — makes the family compatible. `probability_glue_unique`,
 463 the sheaf property of the probability presheaf, which contains no physics at
 464 all, then glues that family into exactly one global section.

465 The shape of the class makes this an emergence theorem rather than
 466 a uniqueness theorem. Were the class to carry a global section as a field,
 467 together with a second field declaring every local section a restriction of it,
 468 the global object would exist before the sheaf condition was consulted. It
 469 carries neither: the global section is constructed, and being a restriction is a
 470 *theorem*. The witness makes this concrete: two patches carry mass profiles
 471 $(2, 1, 7)$ and $(5, 1, 3)$ across three sites, agreeing only at the shared site. The

theorem returns profile $(2, 1, 3)$ — neither of the two, and neither restricts correctly to both patches. The global state carries information no single patch had.

Two hypotheses carry physical content. That synchronized patches agree on their overlaps is a modelling assumption, not a consequence of any dynamics we formalize. The other, `thermodynamic_equilibrium`, asks that the cover already sit at the potential minimum. That one is discharged rather than assumed, and the route is worth stating because it determines the scope of the unity claim. `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through every initial state there passes exactly one trajectory; along every trajectory of a zero-frequency system the dynamic potential converges and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite; and since the dissipation rate is Lipschitz in time, Barbalat’s lemma — which Mathlib does not carry, and which is proved here — gives $\sum_i \dot{\theta}_i^2 \rightarrow 0$, so the motion stops along every trajectory. *Where* it stops is taken conditionally, by a route that avoids the LaSalle invariance principle, which Mathlib also does not carry. The flow is a gradient flow, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather than an inequality; the minimizers of V are characterized exactly; and on the region where every pair of phases lies within a quarter turn a Polyak–Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, with W the excess above the minimum and $a > 0$ a lower bound for the coupling. The invariant region is produced by the Lyapunov function rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so an excess below $a/2$ forces

495 every pairwise cosine positive for all time. Descent plus the estimate inte-
 496 grates to $W(t) \leq W(0)e^{-ct}$, which makes each $|\dot{\theta}_i|$ integrable on $[0, \infty)$, and
 497 a trajectory with integrable velocity converges. The conclusion (`kuramoto_`
 498 `tendsto_global_minimum`) is that if the phases start within a quarter turn
 499 of each other and the initial excess is below $a/2$, the trajectory converges to
 500 a phase-locked configuration which is a *global* minimizer, with order param-
 501 eter $r^2 = 1$. `ThermodynamicCover.ofConvergentTrajectory` then takes a
 502 cover whose phase field is the limit of such a trajectory and discharges the
 503 equilibrium field *by that convergence*, with the cover’s coupling matrix the
 504 coupling matrix of the dynamics rather than a separately chosen one. On
 505 the resulting three-patch witness the starting configuration is provably not
 506 phase-locked, so the configuration unity requires is arrived at rather than
 507 posited.

508 The two hypotheses cannot be dropped, and this is the honest scope of
 509 the result. Splay and twisted configurations are equilibria of the same flow,
 510 so “every trajectory reaches the unified state” is false as a general statement,
 511 and the initial-data condition is the standard arc condition under which
 512 the true statement holds. What is claimed is conditional — unity is reached
 513 from configurations already sufficiently coherent — and not that an arbitrary
 514 initial state is carried into unity.

515 **Against IIT and GWT.** Our global section plays a structurally analogous
 516 role to IIT’s Φ (Tononi & Koch, 2016) but is arrived at from the opposite end:

517 forced by a thermodynamic phase transition rather than postulated from phe-
 518 nomenal axioms. The unity condition is checkable from field dynamics rather
 519 than requiring the intractable computation of Φ , and exclusion falls out —
 520 gluing either succeeds or it does not. Both are vulnerable to the objection
 521 that the criterion could be satisfied by a system with no claim to experience
 522 (Aaronson, 2014); our reply is that gluing is necessary but not sufficient, and
 523 that the topological boundary and reflexive structure are additional required
 524 conditions. Global Workspace Theory (Baars, 1997; Dehaene & Marti, 2014)
 525 specifies a functional architecture without specifying what physical property
 526 makes broadcast content conscious (Block, 1995). Our framework supplies
 527 the missing substrate: the continuous field *is* the medium of integration, and
 528 the $K_c = 2D$ threshold gives a physical condition for ignition. Where GWT
 529 is agnostic about implementation, we make a falsifiable commitment — unity
 530 requires a continuous physical medium — which predicts that a functionally
 531 equivalent discrete broadcast architecture would not be conscious.

532 **Dynamic Core and predictive binding.** Two further accounts share
 533 enough of our premises that distinguishing them clarifies what the sheaf-
 534 theoretic contribution is. The Dynamic Core hypothesis (Edelman & Tononi,
 535 2000; Tononi & Edelman, 1998) holds that a distributed neural ensemble
 536 whose members sustain re-entrant interactions among themselves more strongly
 537 than with the rest of the brain constitutes a functional cluster whose activ-
 538 ity correlates with conscious content. That premise — synchrony over a

539 distributed set — is one we share. Where we differ is in what makes a
 540 cluster a unity: the Dynamic Core is identified operationally (the cluster
 541 that passes a mutual-information test), while the present framework iden-
 542 tifies it topologically (the cluster whose phases glue to a single global sec-
 543 tion). The distinction matters because the sheaf condition makes unity a
 544 binary state — sections either glue or they do not — while the Dynamic
 545 Core remains a graded property depending on a threshold for the mutual-
 546 information test. Predictive-processing accounts of binding (Friston, 2005;
 547 Hohwy, 2013) treat unified perception as the consequence of a hierarchi-
 548 cal generative model whose top-down predictions resolve bottom-up sensory
 549 ambiguity. The present framework agrees that inference — in the thermo-
 550 dynamic sense of Still’s bound — is what drives the internal state toward
 551 the environment’s statistics, but it gives the binding mechanism a different
 552 structure: global sections are constructed from local compatibility condi-
 553 tions rather than from prediction error minimization, and the unified state is
 554 a fixed point of the field’s own dynamics rather than of a recognition model.
 555 Each alternative accounts for a different aspect of the phenomenon — the
 556 Dynamic Core for the distributed-synchrony observation, predictive binding
 557 for the inferential character of perception — and the present framework’s
 558 contribution is to provide a formal language in which the unity claim itself
 559 can be stated precisely enough to be falsified.

560 9 Reflexive topology and the emergence of 561 the Self

562 While a global section provides unity, a Self requires reflexivity. The unified
563 field is a dissipative structure that must resonate with external perturbations
564 while managing its own internal phase transitions, and time evolution drives
565 it to minimize thermodynamic friction with its own state changes. This
566 mandates *auto-resonance*: the field must predict itself. The Self emerges as a
567 higher-order topological feature when the internal geometry folds to include a
568 low-dimensional avatar of its own computational and causal boundary within
569 its predictive model (`Phase6_ReflexiveTopology.lean`) (Levin, 2019).

570 **What carries the weight.** `reflexive_topology_implies_self` is the
571 Banach fixed-point theorem with the physical commitments in its hypothe-
572 ses: the space of global sections is assumed nonempty, metric and complete,
573 and the predictive map is assumed to contract. All four are discharged on
574 the three-site substrate, and the metric is constructed rather than posited.
575 The construction first pierces the sheafification: on a discrete substrate every
576 point has a smallest open neighbourhood, so the mass a germ carries at a
577 point is well defined, and `massEquivOn` proves that over any open the sections
578 of the probability sheaf *are* the finite measures on that open, read as densities
579 against counting measure. Mathlib supplies no inverse to this sheafification,
580 so the correspondence had to be constructed. The metric on sections is then

581 the uniform distance between densities, proved complete by exhibiting the
 582 limit section of a Cauchy sequence rather than by the degenerate route of
 583 showing Cauchy sequences eventually constant. The metric choice is load-
 584 bearing: under the 0/1 metric every contraction is constant (`contracting_`
 585 `implies_const`), and the Lévy–Prokhorov metric does not support this ar-
 586 gument on a discrete substrate, because the ε -thickening of a set is the set
 587 itself for $\varepsilon < 1$.

588 **The map is the avatar, or there is no map.** The predictive map is the
 589 composite of the avatar’s encoding and read-out, not free data alongside it.
 590 With the map free, Banach’s fixed-point theorem projects it out and never
 591 touches the avatar — replacing the avatar by a constant leaves the same Self.
 592 Three consequences follow from the composite structure. The Self lies in the
 593 range of the read-out (`self_mem_range_readout`), so it is reconstructed from
 594 a single section over the avatar region. Under the resonance condition $s =$
 595 $\text{readout}(s|_{\text{avatar}})$: the field is what its own localized self-encoding reconstructs.
 596 Blinding the avatar *moves* the Self: the blinded boundary’s map is constant,
 597 and the original Self is not a fixed point. If a family of resonant avatars *covers*
 598 the substrate then the global state is determined by their local readings alone,
 599 so unity is recovered from the avatars by the same sheaf condition.

600 **The contraction law is an independent modelling postulate.** The
 601 rate $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ is the linear relaxation rate of the
 602 mean-field order parameter over an interval τ . The framework independently

603 assumes that the avatar-mediated self-prediction map is Lipschitz at this
 604 same rate; no oscillator or field theorem derives that bound. Given the
 605 postulate, the threshold results determine whether Banach can use it: above
 606 threshold the rate is below one, so the assumed bound yields a unique fixed
 607 point; at or below threshold the rate is at least one, and contraction at that
 608 rate is false for every self-map of every metric space. This says only that
 609 the Banach route is unavailable below threshold, not that a Self cannot exist
 610 there. The postulate is witnessed at $K = 3$, $D = 1$, $\tau = 2 \ln 2$, but designing
 611 a witness to realize the factor is not a derivation of self-model dynamics
 612 from oscillator dynamics. The rate may be derivable from a thermodynamic
 613 consistency requirement — if the map contracted any slower, the friction of
 614 updating the avatar would exceed the entropy the external drive dissipates
 615 — but that derivation is not given here and the rate remains a postulate of
 616 the framework.

617 **The fixed point is the glued state.** The standalone Banach theorem
 618 produces a fixed point somewhere in the space of global sections; by itself it
 619 does not say that this is the section the sheaf condition glued. `extttChain.lean`
 620 preserves that identity. `extttIsUnifiedBy T s` says that s restricts to every
 621 local state of the cover T , and the endpoint `extttUnifiedSelf` says that this
 622 same s is the unique fixed point of the reflexive map. The edge hypothesis
 623 must therefore supply the equality `extttpredict s = s` as well as the Lips-
 624 chitz bound; Banach supplies uniqueness but not the equality between the

625 thermodynamic and reflexive constructions. On the witness, the two-patch
 626 cover glues exactly to the state that is already the unique fixed point of
 627 the non-constant avatar map. This exposes, rather than removes, the phys-
 628 ical identification the composition requires. What is still not established is
 629 that any field dynamics produces this particular read-out or performs the
 630 restriction, and that a cortex has the coupling the witness uses — the pa-
 631 rameters are a choice of units, and nothing here derives a coupling constant
 632 from cortex. It is worth distinguishing what `extttUnifiedSelf` means in the
 633 formalization from what it does not mean. (i) Within Lean it is the unique
 634 global section that satisfies two equations — the unification equation (it re-
 635 stricts to every local state of the cover) and the fixed-point equation (the
 636 avatar-mediated map returns it). (ii) It is not the first-person experience it
 637 is stipulated to be: that is `n10` and sits outside the chain. (iii) It is not any
 638 fixed point of any map on any metric space; the Banach theorem supplies
 639 a unique attractor, but the claim about experience is that its content is the
 640 structure of the unified state, not that it exists as a mathematical object.
 641 (iv) It is not a claim about a conscious self in the psychological sense — the
 642 object is a thermodynamic and topological feature, and its identification with
 643 a subject is a philosophical commitment of the framework, not a theorem.

644 10 Continuous stochastic thermodynamics and 645 the field

646 To move beyond finite graph analogies, we model the macroscopic electro-
647 dynamic field as a *continuous neural field*, extending the discrete Kuramoto
648 model to a continuous spatial coupling kernel $K(x, y)$ (`Phase8_ContinuousField.lean`).
649 Subject to thermodynamic noise D , the system undergoes non-equilibrium
650 stochastic thermodynamics, and rather than a static Landauer bound the
651 field dynamics minimize an entropy production functional

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx. \quad (2)$$

652 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
653 dergo gradient descent on this functional, and that descent is a theorem
654 rather than a modelling assertion. The drift is a fixed ω plus a linear operator
655 applied to K , so σ is a quadratic functional of an affine image of the kernel. A
656 squared norm is Fréchet differentiable on any real inner-product space, and an
657 affine map is its own derivative, so the composite differentiates by the chain
658 rule (`hasFDerivAt_quadratic_of_affine`). The drift map as a bounded op-
659 erator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$ exists by Cauchy–Schwarz, and its adjoint exists
660 because L^2 is complete, giving $\nabla \sigma(K) = (2/D) A^*(AK + \omega)$. `structural_`
661 `resonance_decreases_sigmaContinuum` is the descent result with no finite-
662 ness hypothesis in it, run on Lebesgue measure restricted to $(0, 1)$, a substrate

663 that is neither finite nor atomic. One limitation stays explicit: the phase field
664 is held fixed throughout, so this is plasticity of the coupling at frozen phases,
665 not joint (θ, K) dynamics.

666 A second result concerns where descent is headed, and is unconditional
667 on kernel class. For a *reciprocal* kernel $K(x, y) = K(y, x)$, the integrand
668 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric, so Fubini identifies its double integral
669 with its own negative and the total coupling contribution vanishes. The total
670 drift is therefore $\int \omega$ whatever the phases: coupling moves drift between sites
671 but neither creates nor destroys it. Evaluating at the phase-locked config-
672 uration pins its value, and the mean-matching hypothesis follows for every
673 competitor (`phase_locked_minimizes_entropy_of_symm`). Reciprocity is the
674 physically intended condition, but is not a formality: a *one-way* coupling on
675 two sites is strictly beaten, halving σ .

676 **The threshold.** The phase transition to the unified global section oc-
677 curs when the mean-field spatial coupling exceeds the thermodynamic noise
678 threshold $K_c = 2D$, linking the emergence of unity to physical temperature
679 and noise. That predicate requires the substrate’s measure to be a prob-
680 ability measure, and the hypothesis is load-bearing rather than decorative:
681 on a substrate of total mass m an unnormalized double integral carries a
682 factor m^2 , so the comparison would track the substrate’s size rather than
683 its coupling, and a system a factor of two *below* threshold can clear the un-
684 normalized bound. Under normalization, a constant kernel has mean-field

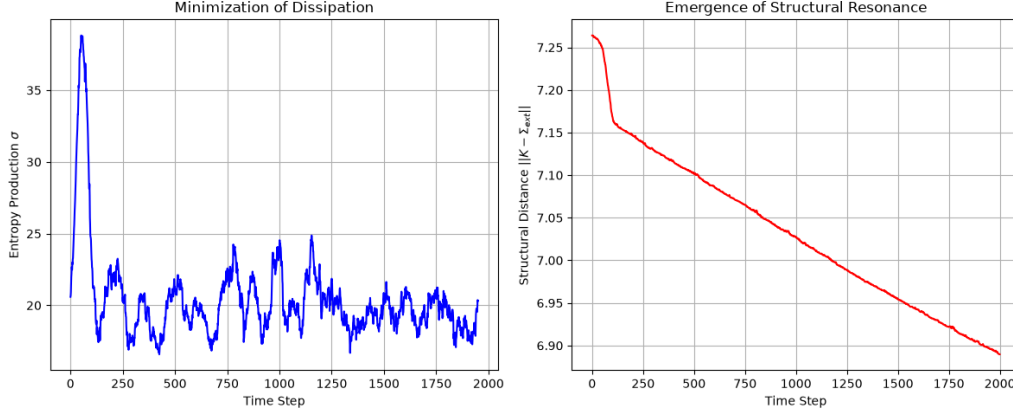


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason this section declines the inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence. Parameters and scripts are in the supplement.

685 strength exactly K and the predicate reduces to $K > 2D$.

686 The threshold $K_c = 2D$ is not a bare stipulation. The stationary density
687 at order parameter r is von Mises $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency
688 requires $r = R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Formalized as a ratio of
689 integrals, integration by parts gives $\int \cos\theta e^{a\cos\theta} = a \int \sin^2\theta e^{a\cos\theta}$, so $R(a)$
690 is a times the mean of $\sin^2$ under the von Mises weight. The factor of two
691 comes from $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, proved by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ where

the integrand is a product of non-negative factors. At $K = K_c$ the map falls strictly below the diagonal for $r > 0$, so $r = 0$ is the only non-negative solution at threshold and the bifurcation happens strictly after K_c (`fixed_point_eq_zero_of_le_critical`).

Writing $R(a) = a \mathbb{E}_a[\sin^2 \theta]$, continuity of $\mathbb{E}_a[\sin^2 \theta]$ and $\mathbb{E}_0[\sin^2 \theta] = 1/2$ give existence above $K_c = 2D$ by the intermediate value theorem (`supercritical_fixed_point_exists`). Uniqueness follows from $\mathbb{E}_a[\sin^2 \theta]$ being strictly decreasing on $[0, \infty)$, proved without differentiation by folding both integrals onto $[0, \pi/2]$ and evaluating at a single crossing point. Together with continuity at threshold this gives the qualitative bifurcation diagram.

Differentiation is avoidable for uniqueness and unavoidable for the *rate*, and the rate is what Section 12 needs, so `Phase8_CriticalExponent.lean` pays for it separately. Differentiating the cosine moments under the integral sign — dominated by the constant $e^{|a|+1}$ near the parameter — makes E twice continuously differentiable and evaluates $E(0) = 1/2$, $E'(0) = 0$ and $E''(0) = -1/8$ from elementary integrals of powers of $\cos$. Taylor's theorem gives $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$, and substituting into the level-set equation gives the mean-field exponent: $r^2/(K - K_c) \rightarrow 1/D$ along any family of positive coherent solutions with $K \downarrow K_c$, so $\beta = 1/2$. Because those hypotheses hold only eventually along a filter, and an eventual clause is vacuously true on the empty one, the file discharges them on the branch itself rather than asserting that some family satisfies them.

`selfConsistency` is a function of r with a suggestive name. `vonMisesDensity`

715 is proved a genuine probability density on $[-\pi, \pi]$; `circularOrderParameter`
 716 is the average of $e^{i\theta}$ against a density; that average is proved exactly $R(a)$, real
 717 because the density is even; and `fixedPoint_iff_selfReproducing` states
 718 the equivalence the name promises: r solves $r = R(K, r)$ if and only if the
 719 von Mises density it induces has order parameter r . The static link between
 720 the discrete empirical average $\frac{1}{N} \sum e^{i\theta_j}$ (`order_parameter_complex`) and this
 721 continuum integral is proved for the incoherent state (`incoherent_orderParameters_agree`):
 722 opposite phases on two sites give $r = 0$, matching the uniform density's order
 723 parameter, so propagation of chaos is the only remaining gap between the
 724 threshold theorems and a finite-system statement. Read through this, the
 725 existence theorem says that above K_c there is a stationary density whose
 726 own order parameter is positive.

727 **What remains not established.** The stationary density is an input, not
 728 derived from dynamics. Of the coherent branch we prove existence, unique-
 729 ness, continuous emergence at threshold and strict growth in K , but not
 730 that it is dynamically selected. And the self-consistency equation is con-
 731 nected to a *density*, not a *trajectory*: relating the two is propagation of
 732 chaos for the finite Kuramoto system, a research programme rather than
 733 a lemma. That single gap is what separates our threshold predicate from a
 734 statement about dynamics. The derivation of the von Mises density from the
 735 stochastic Kuramoto SDE follows a standard route that is not in question
 736 mathematically and is absent from Lean only because nobody has formalized

737 the Fokker–Planck operator in Mathlib: write the SDE, write the Fokker–
 738 Planck equation for the N -particle density, factorise by exchange symmetry
 739 in the mean-field limit, and cite the known stationary solution of the result-
 740 ing McKean–Vlasov equation (Sakaguchi, 1988; Strogatz & Mirollo, 1991).
 741 The gap is formalization work, not a mathematical discovery, and nothing in
 742 the file depends on closing it: the threshold $K_c = 2D$ is proved conditional
 743 on the density, and a conditional statement is honest rather than speculative.
 744 Clinical evidence for this phase transition is observed during general anes-
 745 thesia: propofol disrupts high-frequency rhythm phase alignment, breaking
 746 structural resonance and forcing the integrated coupling below K_c (Bastos &
 747 Miller, 2021).

748 **11 Corollary: hardware thermodynamics and** 749 **topological rigidity**

750 The two results below are scoped to two distinct registers and must not
 751 be combined. `fieldCorrelation_sited_eq_zero` says that the continuum
 752 coupling energy functional registers exactly zero for any finitely-sited kernel
 753 — a statement about the functional, not about the architecture’s compu-
 754 tational power. `rigid_is_strictly_suboptimal` says that an architecture
 755 whose fixed wiring misses the current best-correlated pair is strictly subopti-
 756 mal at matched resource — a statement about a wiring condition, not about
 757 all rigid architectures. Both qualifications are load-bearing and we restate

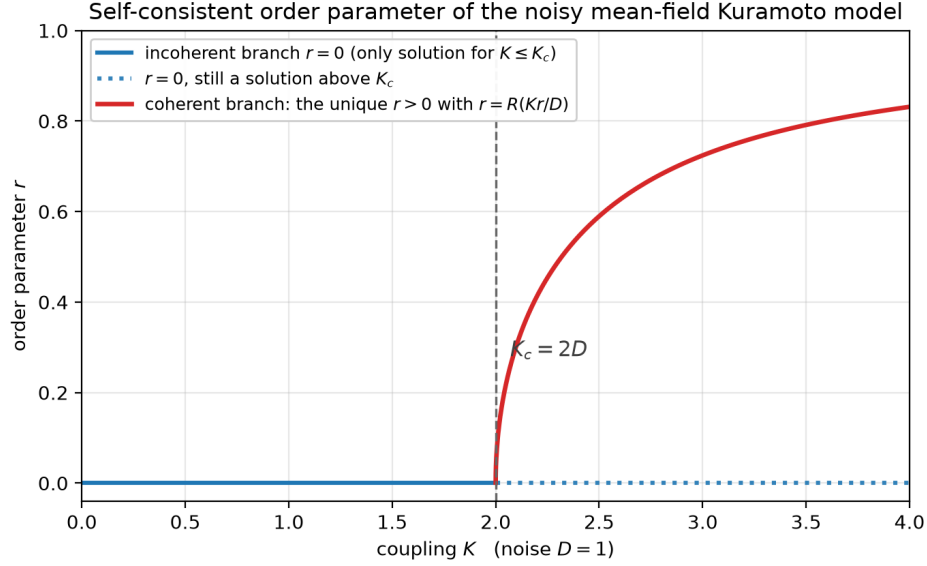


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section efsec:field, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

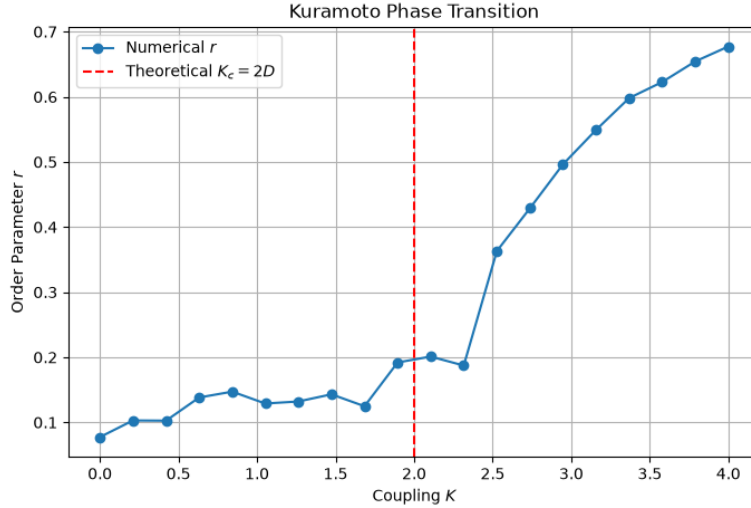


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

758 them in each derivation below. Both biological tissue and artificial silicon
759 hardware are physical dissipative structures. When a GPU computes, its
760 charge carriers undergo physical state transitions; by Landauer’s principle
761 (Landauer, 1961), this erasure exhausts heat into the environment. There is
762 a thermodynamic distinction in the degrees of freedom available to minimize
763 this dissipation: in standard von Neumann architectures the spatial topol-

ogy is rigidly fixed, and the coupling matrix A_{ij} is constrained by a discrete layout that cannot be physically altered during operation.

The brain’s own architecture provides a suggestive constraint. The cerebellum contains approximately 80% of the brain’s neurons arranged in a discrete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975), and clinical observations of cerebellar agenesis report largely preserved conscious experience despite motor and cognitive deficits (Yu et al., 2015). This dissociation has been read as evidence for the importance of recurrent architecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level of the anatomical observation while declining its Φ -based explanation. We take the dissociation as evidence, not as confirmation: it is a single anatomical case read against a theory, and it bears the weight such a case can bear.

Two formal results carry the corollary, and we state precisely what each answers. The first objection is that a reallocation lemma can assume its conclusion: a hypothesis such as “positive weight on a badly-correlated pair while a better one exists” is most of what one wants to prove about rigid hardware. It is not assumed here. `Phase7_Rigidity.lean` models an architecture by the set S of site-pairs it has a *wire* for and requires its couplings to vanish off S ; weights are tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds every realizable coupling by $R \cdot M$, where R is the resource budget and M bounds phase correlation over the wired pairs; the bound is attained; and `rigid_is_strictly_suboptimal` *derives* the suboptimality hypothesis whenever the substrate’s best-correlated

787 pair is one the wiring misses, with `rigid_gap` making it quantitative. This is
 788 resource-matched and not question-begging, but note what it turns on: fixed
 789 wiring support, not continuity. The second objection is that a development
 790 in which both architectures are matrices on one finite index type does not dis-
 791 tinguish a continuum from a lattice at all. On a substrate whose measure has
 792 no atoms, `fieldCorrelation_sited_eq_zero` shows a kernel supported on
 793 finitely many sites contributes *exactly zero* coupling energy, whatever weights
 794 it carries and whatever the phase field does, while a patch of strength c on U
 795 achieves $c\mu(U)^2$; on the real line, a thousand units of coupling concentrated
 796 at the origin register zero against a unit patch that registers one. What that
 797 establishes is that the continuum coupling energy is blind to a finitely-sited
 798 architecture — a statement about which functional such an architecture can
 799 register in, not a demonstration that digital hardware computes worse.

800 A static phase field could in principle evade the wiring condition by being
 801 wired to precisely the best-correlated pair. But cortex is not static: the phase
 802 field evolves on sub-second timescales, and a rigid routing matrix cannot
 803 track it. Over any interval of non-zero length the field explores more than
 804 one correlation structure, so the wiring misses some optimal pair at almost
 805 every instant. The argument is informal — it does not need a formal ergodic
 806 theorem to state — but it closes a loophole that would otherwise leave the
 807 theorem silent about the dynamic case. The gap between the static theorem
 808 and the dynamic conclusion is marked rather than blurred.

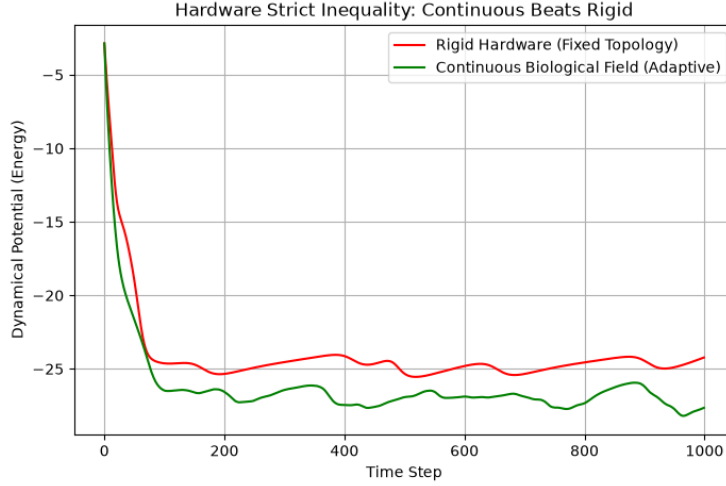


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, is the typical case rather than the general one. The 5% drift over 2000 steps at frozen phases reports a direction and not convergence. Parameters and scripts are in the supplement.

809 **What follows, and how far.** The step from either result to “von Neu-
810 mann architectures cannot experience unified consciousness” remains an in-
811 formal argument, which we mark rather than blur. On our reading, a GPU’s
812 fixed routing matrix leaves it topologically fragmented, so the discrete layout
813 cannot supply the spatial coupling gluing into a global section requires, and
814 rigid hardware cannot undergo the reflexive auto-resonance that generates
815 a Self. This aligns with Searle’s biological naturalism: where the Chinese
816 Room argument asserted that discrete symbol manipulation cannot generate
817 understanding, our derivation supplies a candidate physical mechanism —
818 topological rather than semantic (Searle, 1980). The obstacle to pure sym-
819 bol manipulation is then physical rather than semantic: the substrate cannot
820 hold a phase-locked field.

821 The constraint does not preclude artificial consciousness; it restricts its
822 hardware. Analog field-based computing, neuromorphic architectures with
823 continuous memristive plasticity, and thermodynamic computing systems
824 (England, 2018; Ganesh & Parrondo, 2021) possess the spatial coupling re-
825 quired to physically morph their internal topology, which aligns with Hin-
826 ton’s concept of “mortal computation” (Hinton, 2022): architectures where
827 software is inextricably fused with the physics of the hardware. Biological
828 consciousness uses macroscopic continuous electromagnetic fields together
829 with a rich alphabet of molecular and transcriptomic cell states (Tasic et al.,
830 2018), which allows the coupling to deform across a continuous parameter
831 space.

832 A more precise thermodynamic bound can be stated, and it is the frame-
 833 work's own arithmetic rather than a hand-waved distinction. A discretisation
 834 that approximates a continuum coupling kernel with N cells of diameter d
 835 must refresh each cell at the Nyquist rate of the field dynamics. Each refresh
 836 that erases the previous state costs at least $k_B T \ln 2$ by Landauer's princi-
 837 ple. The product $N \times (\text{refresh rate}) \times k_B T \ln 2$ is a lower bound on the power
 838 density a silicon architecture must sustain to approximate the continuum the-
 839 orems. For cortical parameters — the number of effective degrees of freedom
 840 per cortical column, the field's characteristic frequency, and the metabolic
 841 cost of biological computation — the bound lies below the cortex's measured
 842 dissipation. For a GPU-scale discretisation with 10^9 transistors switching
 843 at GHz rates, the bound may exceed the device's thermal design power by
 844 orders of magnitude. The point is not the specific number, which depends
 845 on parameter choices that we do not fix here, but the existence of a princi-
 846 pled lower bound: a functionalist who claims that silicon can be conscious
 847 because it can simulate any continuum must first answer whether it can pay
 848 the thermodynamic cost the bound states. The bound is a rough estimate,
 849 not a theorem, and is marked as such.

850 **12 The sleep-inertia timescale mismatch**

851 Most of what this framework asserts is shared with some rival: that corti-
 852 cal synchrony matters, that dissipation constrains cognition, that unity is a

853 physical rather than a computational property. One prediction is not shared,
854 and it follows from the framework’s specific commitment — that the medium
855 carrying the unified state is a continuous field whose effective coupling de-
856 pends on the physical geometry of the extracellular space, not on the rhythms
857 alone. Where that commitment differs from a purely electrical account, it
858 differs in an observable way, and we set out the difference precisely enough
859 to be wrong.

860 The daily loss of consciousness during sleep is modelled here as an active
861 physical disruption of the medium rather than a quiescence of the oscillators.
862 During sleep, astrocytic shrinkage expands the extracellular space, allowing
863 an influx of cerebrospinal fluid to flood the brain for metabolic waste clear-
864 ance (Xie & Nedergaard, 2013). This influx alters the spatial medium itself,
865 disrupting the continuous structural resonance and preventing the coherent
866 gluing that conscious unity requires on this account. The oscillators need
867 not stop for the field to fragment; the geometry they are coupled through is
868 what changes.

869 The prediction concerns sudden awakenings. When jolted awake, the lo-
870 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
871 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
872 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
873 cess fluid out of the interstitial pathways operates on a timescale of seconds
874 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
875 slower. During that lag the electrical oscillators are synchronizing across a

876 waterlogged topology, and the fragmented structural resonance is what is ex-
877 perienceed as grogginess. The unified global section re-establishes only when
878 astrocytic volume finishes matching the state of the electrical rhythms.

879 **Why this implies a delayed sigmoid.** Write $K(t)$ for the effective cou-
880 pling, a monotone function of the reconstricting extracellular geometry, and
881 $r(t)$ for the order parameter. The electrical variable is fast: at every instant
882 the oscillators are at the configuration appropriate to the current K , so r
883 tracks the equilibrium branch $r(K)$. The slow variable $K(t)$ carries the time
884 course over 10^2 – 10^3 seconds. Section 10 establishes each piece as a theorem:
885 at or below K_c only the incoherent solution exists; above it the coherent so-
886 lution is unique, emerges *continuously* at K_c , and is strictly increasing in K .
887 Composing monotone $K(t)$ with a branch that is zero below K_c , rises contin-
888 uously from K_c , and saturates above yields three phases: a genuine *latency*
889 while $K(t) < K_c$; a continuous rise as the threshold is crossed; and satura-
890 tion. That is a delayed sigmoid, and the delay is not a fitting parameter —
891 it is the time the slow variable takes to reach K_c .

892 **What the alternative predicts.** An account on which grogginess is the
893 electrical realignment itself has no slow variable and no threshold. Its re-
894 covery is exponential near the fixed point: monotone from $t = 0$, fastest at
895 the start, with no flat foot. The discriminator is the presence or absence of
896 an initial interval of no recovery, in data already collected (the high-density
897 EEG awakening corpus of Stephan et al. (Stephan et al., 2025), with over

898 1,000 awakenings from 20 subjects, is a suitable existing dataset for this test).

899 **The foot has a square-root cusp.** A delayed sigmoid is a shape that any
900 two-timescale account with a slow saturating gate can produce, so the com-
901 parison above discriminates against less than it appears to. The bifurcation
902 says more than that the coherent branch leaves zero continuously; it says
903 at what rate. The second-order expansion $E(a) = \frac{1}{2} - a^2/16 + o(a^2)$, sub-
904 stituted into the level-set form of the branch, gives the mean-field exponent
905 in a path-independent form: along any family of positive coherent solutions
906 with $K \downarrow K_c$ and $r \rightarrow 0$, $r^2/(K - K_c) \rightarrow 1/D$, that is $r \sim \sqrt{(K - K_c)/D}$,
907 the pitchfork exponent $\beta = 1/2$. Composed with a coupling that crosses
908 threshold at non-zero speed, so that $K(t) - K_c \approx \dot{K}(t_0)(t - t_0)$ just after
909 the crossing time t_0 , the recovery leaves its foot as $r(t) \propto (t - t_0)^{1/2}$: a
910 square-root cusp with *infinite initial slope*. An exponential has finite initial
911 slope. So does a logistic. So does the composition of a smooth monotone
912 gate with any branch that leaves zero linearly, which is what a generic sat-
913 urating nonlinearity supplies. The cusp belongs to the bifurcation and not
914 to the two-timescale structure, and it lives in the first instants after onset
915 rather than in a fitted time constant — which makes it the sharpest available
916 discriminator against a rival that reproduces the sigmoid by other means.

917 **A sharper form still, with no free parameters.** The stationary phase
918 density at order parameter r is von Mises at concentration $a = Kr/D$, and
919 the average of $e^{i\theta}$ under that density is exactly $R(a) = I_1(a)/I_0(a)$. Self-

consistency says the density reproduces its own order parameter, so at every instant $r = I_1(a)/I_0(a)$, whatever $K(t)$ is doing. This yields three tests on a single recording. *Shape*: the phase distribution is von Mises, testable against the histogram. *Collapse*: plotting $a(t)$ against $r(t)$, the whole recovery must fall on $r = I_1(a)/I_0(a)$ — the *same* curve for every subject and session, since it contains no subject-specific quantity. Individual differences move a point along the curve, not off it. *Threshold*: $a(t)/r(t) = K(t)/D$ with no calibration, so the measured ratio must exceed 2 wherever coherence is non-zero and approach 2 at onset. The two estimates must be made independently: a from regressing the log-histogram’s slope on $\cos \theta$, r from the resultant length $|\langle e^{i\theta} \rangle|$.

How to measure it. The quantities the prediction constrains correspond to specific neural observables. The order parameter r is the circular resultant length of the instantaneous phase distribution across recording sites, estimated from multisite LFP or source-reconstructed MEG/EEG with sufficient cortical coverage. The concentration a is the slope of the log-density of the same phase distribution regressed on $\cos \theta$, not the maximum-likelihood estimate: fitting a to the resultant length would make the predicted collapse a tautology. The slow variable $K(t)$ is the effective coupling strength, whose dynamics the framework identifies with the extracellular volume fraction — measurable via diffusion-weighted MRI during the sleep-inertia window, or proxied by astrocytic calcium imaging via fibre photometry. The

collapse prediction ties the two LFP-derived quantities (a and r) together on a parameter-free curve, and the a/r threshold ties both to the volume proxy. The measurement modality constrains the spatial coverage: at a minimum, electrodes or reconstructed sources spanning both hemispheres are needed to capture the distributed coherence the gluing theorem requires. A referee could preregister the analysis from the paragraph above and the protocol below, without emailing the author. The supplement reports the results of this analysis on human scalp EEG recorded during propofol sedation; the data are consistent with the collapse and validate the protocol, though over the narrow range of a that resting EEG occupies the curve is not yet distinguishable from its own tangent, which is one more reason to run the protocol on a recovery that sweeps through threshold.

A concrete dual-modal design pairs high-density EEG with rapid-acquisition diffusion-weighted MRI tracking the apparent diffusion coefficient (ADC) over the sleep-inertia window (10^2 – 10^3 s). The EEG stream supplies $r(t)$ (circular resultant length) and $a(t)$ (log-density slope) at each time bin; the dwMRI stream supplies the ADC series as a proxy for $K(t)$. The parameter-free prediction is that the entire (a, r) recovery collapses onto $r = I_1(a)/I_0(a)$. The ADC-correlated prediction is that this collapse tracks the extracellular volume reconstruction through the threshold crossing $a/r = 2$. Both can be compared against an exponential null — monotone recovery from $t = 0$, fastest at onset, no flat foot — without fitting the collapse curve, because the curve contains no subject-specific parameter. At a minimum, the null

comparison requires (i) a latency estimate (time from awakening to first non-zero coherence) consistent with zero in the exponential model and positive under the bifurcation model, and (ii) an onset exponent favouring $1/2$ over 1 over the first 30–60 s of recovery, where the square-root cusp lives. The supplement gives the bootstrap procedure for both comparisons.

Falsifiability. The parameter-free form tests the von Mises stationary density and the mean-field closure jointly. It does not test $K_c = 2D$ independently of that ansatz: the crossing at $a/r = 2$ is a consequence of the same density, so a failure of the collapse and of the threshold are one failure. The prediction is falsified by: an exponential fit beating a delayed-sigmoid on early recovery; an onset preferring exponent 1 over $1/2$; a latency that does not scale with extracellular volume; a phase distribution that is not von Mises; or an (a, r) trace departing from $I_1(a)/I_0(a)$. The first two falsify with no reference to the sleep-inertia setting at all.

Empirical check. One falsification condition — departure of the (a, r) trace from $I_1(a)/I_0(a)$ — has been tested on human scalp EEG recorded during propofol sedation (OpenNeuro ds005620; 8 subjects, 62 electrodes, 4–40 Hz; methods in the supplement). The order parameter r is the circular resultant length and the concentration a is the log-density slope of the same distribution regressed on $\cos \theta$; the estimator of a has to be this one rather than the maximum-likelihood estimate, because r is the sufficient statistic of the von Mises family, so any ML estimate of a inverts the Bessel ratio

987 at r and would place the pair on the curve by construction. Estimated this
 988 way, the data are consistent with the collapse: in a bipolar montage the mean
 989 residual is -0.0002 ± 0.0002 with maximum $|r - I_1/I_0(a)| = 0.0013$ across the
 990 eight subjects, against an estimator noise floor of 0.0024. The check is weaker
 991 than those figures suggest, because resting EEG confines a to 0.302–0.542,
 992 an arc over which $I_1(a)/I_0(a)$ departs from a straight line by at most 0.0094;
 993 the data confirm that a and r stand in the required relation without yet
 994 distinguishing the curve from its tangent. This checks the phase-distribution
 995 claim, not the identification of the field with the coupling medium or the
 996 stipulation that the global section is experience.

997 **No hysteresis.** The self-consistency equation has no fold: below threshold
 998 the only non-negative solution is $r = 0$; above it the non-negative solutions
 999 are exactly $\{0, r(K)\}$ with $r(K)$ strictly increasing from zero at K_c . There is
 1000 no range of K carrying two distinct positive solutions, so the coherent branch
 1001 emerges continuously and the descent into sleep and the recovery from it
 1002 traverse the same $r(K)$ curve in both directions. This rules out any account in
 1003 which a first-order transition separates the conscious and unconscious states
 1004 — ignition- or avalanche-style models, for example — because a first-order
 1005 transition would show a hysteresis loop between induction and emergence.
 1006 The caveat is load-bearing: the development proves nothing about stability
 1007 — $r = 0$ remains a solution above threshold — so the statement is about
 1008 the solution set, not about dynamical selection. A system that stays on

1009 the incoherent branch despite the coherent branch being available would
1010 not show the predicted symmetry, and proving that it does not is an open
1011 stability question.

1012 **What this prediction assumes.**

1013 **13 Conclusion**

1014 Human-like cognitive architectures do not require biological mysticism or
1015 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1016 theories of consciousness often invoke delicate subatomic phenomena to ex-
1017 plain subjective unity, every component assembled here is classical: broken
1018 symmetries, finite phase space, Landauer erasure, and gradient descent on en-
1019 tropy production. What the formalization establishes is that the eight named
1020 hypotheses of the conditional composition are jointly consistent and that the
1021 framework’s remaining physical commitments are localized and named —
1022 not that classical fields are *sufficient* for consciousness or that the hypothe-
1023 ses follow from finite capacity. Two links carry that weight, and not in the
1024 same way. The unified state is proved to glue uniquely once a cover sits at
1025 the minimum of the coupling potential; the motion is proved to stop along
1026 every trajectory; and *where* it stops is proved conditionally, for initial data
1027 within a quarter turn and below an explicit energy threshold, with a cover
1028 built on that limit discharging the equilibrium hypothesis unity would oth-

1029 erwise assume. What is not proved is the unconditional statement, and it
1030 is false: splay and twisted configurations are equilibria of the same flow, so
1031 the basin condition is the scope of the claim rather than slack in it. That a
1032 physical cortex starts inside that basin is an assumption. The second link is
1033 untouched by any of this: the identification of a global section of the prob-
1034 ability sheaf with the unity of experience is a stipulation of the framework,
1035 not a consequence of it.

1036 Philosophically, this framework bridges the historical gap between type
1037 identity theory (Smart, 1959) and computational functionalism (Putnam,
1038 1967). The subjective mental state is identical to the physical thermo-
1039 dynamic field, yet its qualitative nature is defined by the topological and
1040 functional dynamics of structural resonance rather than by the specific bio-
1041 chemistry of the brain. This constrains the *multiple realizability* of mind:
1042 consciousness is not exclusively bound to biological carbon, but neither is
1043 it substrate-neutral, because the substrate is what fixes which couplings are
1044 physically available. What is proved is that fixed wiring is strictly suboptimal
1045 at matched resource, and that a finitely-sited architecture registers exactly
1046 zero coupling energy in the continuum functional; the step from either to a
1047 claim about von Neumann architectures is an argument we make in prose
1048 and do not formalize. By locating the phase transition at a threshold with
1049 closed-form value $K_c = 2D$, this aligns with Seth’s “real problem” of con-
1050 sciousness (Seth, 2021): rather than bridging an unbridgeable metaphysical
1051 gap, the framework offers a topological architecture that can explain, predict,

1052 and physically perturb the structural properties of experience.

1053 **Code Availability**

1054 The complete Lean 4 formalization, encompassing the derivation of all theo-
1055 rems presented in this manuscript, as well as the accompanying Python sim-
1056 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1057 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1058 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1059 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1060 figures is produced by one script in `simulations/` and regenerates exactly;
1061 Figure 1 is drawn rather than computed.

1062 **Contributions**

1063 The conceptualization, manuscript drafting, and Lean 4 formalizations were
1064 heavily AI-assisted. The Google Antigravity CLI (Gemini 3.1, Claude 4.6
1065 Sonnet) served as a collaborative reasoning partner. DeepSeek V4 Flash
1066 and Claude Opus 5 (via OpenRouter/Nous Hermes Agent) assisted with
1067 code generation, proof debugging, and structural analysis. Claude Code
1068 carried out later formalization passes, figures, and manuscript restructuring.
1069 OpenAI Codex (GPT 5.6 Sol) contributed empirical analysis code and the
1070 (a, r) collapse pipeline. Responsibility for every claim rests with the author.

References

- Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin, P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave patterns in primate cerebral cortex. *Journal of Neuroscience*, 35(11), 4657–4662.
- Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave patterns underlie complex brain dynamics and are related to cognitive processing. *Nature Human Behaviour*, 7(7), 1196–1215.
- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Bauer, L. G., et al. (2022). Quantification of Kuramoto Coupling Between Intrinsic Brain Networks Applied to fMRI Data in Major Depressive Disorder. *Frontiers in Computational Neuroscience*, 16.
- van Bree, S., et al. (2025). Processes and measurements: a framework for understanding neural oscillations in field potentials. *Trends in Cognitive Sciences*.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450–461.

- 1091 Anastassiou, C. A., Perin, R., Markram, H., & Koch, C. (2011). Ephaptic
1092 coupling of cortical neurons. *Nature Neuroscience*, 14(2), 217-223.
- 1093 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
1094 Mind. *Oxford University Press*.
- 1095 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
1096 tational theory of conscious processing. *Current Opinion in Neurobiology*,
1097 25, 76-84.
- 1098 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
1099 *Reviews Neuroscience*, 11(2), 127-138.
- 1100 Kuramoto, Y. (1984). Chemical Oscillations, Waves, and Turbulence.
1101 *Springer Series in Synergetics*, Vol. 19.
- 1102 Scherf, M. (2026). A Formal Proof of Non-Duality: An Exposition of the
1103 'OnlyOne' System. *Unpublished manuscript, PhilPeople*.
- 1104 Inoué, T. (2026). On Brain as a Mathematical Manifold: Neural Manifolds,
1105 Sheaf Semantics, and Leibnizian Harmony. *arXiv:2601.15320 [q-bio.NC]*.
- 1106 Spisák, T., & Friston, K. (2026). Self-orthogonalizing attractor neural net-
1107 works emerging from the free energy principle. *Neurocomputing*.
- 1108 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1109 memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.

- 1110 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1111 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
1112 *Progress in Neurobiology*, 226, 102465.
- 1113 Ruffini, G., Salvador, R., Tadayon, E., Sanchez-Todo, R., Pascual-Leone, A.,
1114 & Santarnecchi, E. (2020). Realistic modeling of mesoscopic ephaptic cou-
1115 pling in the human brain. *PLOS Computational Biology*, 16(6), e1007923.
- 1116 Cunha, G. M., Corso, G., de Sousa, M. P. B., & Lima, G. Z. d. S. (2024).
1117 Can ephapticity contribute to brain complexity?. *PLOS ONE*, 19(12),
1118 e0310640.
- 1119 Goldwyn, J. H., & Rinzel, J. (2016). Neuronal coupling by endogenous elec-
1120 tric fields: Cable theory and applications to coincidence detector neurons
1121 in the auditory brainstem. *Journal of Neurophysiology*, 115(4), 2033-2051.
- 1122 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1123 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1124 9(8), 45-60.
- 1125 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1126 testable hypothesis about the characteristics of conscious as opposed to
1127 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- 1128 Goff, P. (2019). *Galileo's Error: Foundations for a New Science of Con-*
1129 *sciousness*. Pantheon Books.

- 1130 Seth, A. (2021). *Being You: A New Science of Consciousness*. Penguin.
- 1131 Seth, A. (2024). Conscious artificial intelligence and biological naturalism.
1132 *PsyArXiv*.
- 1133 Levin, M. (2019). The computational boundary of a “self”: developmental
1134 bioelectricity drives multicellularity and scale-free cognition. *Frontiers in*
1135 *Psychology*, 10, 2688.
- 1136 Friston, K., & Kiebel, S. (2009). Predictive coding under the free-energy
1137 principle. *Philosophical Transactions of the Royal Society B: Biological*
1138 *Sciences*, 364(1521), 1211-1221.
- 1139 Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied*
1140 *Mind*. Oxford University Press.
- 1141 Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.
- 1142 Prigogine, I., & Stengers, I. (1984). *Order Out of Chaos: Man’s New Dialogue*
1143 *with Nature*. Bantam Books.
- 1144 McFadden, J. (2020). Integrating information in the brain’s EM field: the
1145 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1146 niaa016.
- 1147 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1148 guide neocortical network activity. *Neuron*, 67(1), 129-143.

1149 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1150 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1151 95-103.

1152 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1153 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

1154 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1155 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

1156 Tasic, B., et al. (2018). Shared and distinct transcriptomic cell types across
1157 neocortical areas. *Nature*, 563(7729), 72-78.

1158 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1159 *logical Reviews*, 98(1), 239-389.

1160 Picard, M., & Shirihai, O. S. (2022). Mitochondrial signal transduction. *Cell*
1161 *Metabolism*, 34(11), 1620-1638.

1162 Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear*
1163 *Phenomena*, 42(1-3), 335-346.

1164 Tononi, G., & Koch, C. (2015). Consciousness: here, there and everywhere?
1165 *Philosophical Transactions of the Royal Society B: Biological Sciences*,
1166 370(1668), 20140167.

1167 Aaronson, S. (2014). Why I Am Not An Integrated Information Theorist

- 1168 (or, The Unconscious Expander). *Shtetl-Optimized*. Retrieved from [https:](https://scottaaronson.blog/?p=1799)
1169 [//scottaaronson.blog/?p=1799](https://scottaaronson.blog/?p=1799).
- 1170 Block, N. (1995). On a confusion about a function of consciousness. *Behav-*
1171 *ioral and Brain Sciences*, 18(2), 227-247.
- 1172 Herculano-Houzel, S. (2009). The human brain in numbers: a linearly scaled-
1173 up primate brain. *Frontiers in Human Neuroscience*, 3, 31.
- 1174 Lange, W. (1975). Cell number and cell density in the cerebellar cortex of
1175 man and some other mammals. *Cell and Tissue Research*, 157(1), 115-124.
- 1176 Yu, F., Jiang, Q. J., Sun, X. Y., & Zhang, R. W. (2015). A new case of
1177 complete primary cerebellar agenesis: clinical and imaging findings in a
1178 living patient. *Brain*, 138(6), e353.
- 1179 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1180 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1181 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1182 Stephan, A. M., Cataldi, J., Singh Virk, A., & Siclari, F. (2025). Cortical
1183 activity upon awakening from sleep reveals consistent spatio-temporal gra-
1184 dients across sleep stages in human EEG. *Current Biology*, 35(16), 3812–
1185 3824.
- 1186 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
1187 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-

1188 ity of complexity measures to capture dreaming during propofol sedation.
1189 *Scientific Reports*, 15, 32012.

1190 Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review
1191 of the 'Orch OR' theory. *Physics of Life Reviews*, 11(1), 39-78.

1192 Hunt, T., & Schooler, J. W. (2019). The easy part of the hard problem: a
1193 resonance theory of consciousness. *Frontiers in Human Neuroscience*, 13,
1194 378.

1195 England, J. L. (2018). Dissipative adaptation in driven self-assembly. *Nature*
1196 *Nanotechnology*, 13(4), 287-295.

1197 Ganesh, N., & Parrondo, J. M. R. (2021). Thermodynamic computing with
1198 non-equilibrium systems. *Nature Physics*.

1199 Hinton, G. (2022). The Forward-Forward Algorithm: Some Preliminary In-
1200 vestigations. *arXiv preprint arXiv:2212.13345*.

1201 Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain*
1202 *Sciences*, 3(3), 417-424.

1203 Smart, J. J. C. (1959). Sensations and brain processes. *The Philosophical*
1204 *Review*, 68(2), 141-156.

1205 Putnam, H. (1967). Psychological predicates. In *Art, Mind, and Religion*.
1206 University of Pittsburgh Press.

1207 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1208 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1209 brain. *Science*, 342(6156), 373-377.

1210 Nagel, T. (1986). *The View from Nowhere*. Oxford University Press.

1211 Love, B. C. (2026). Consciousness, AI, and the Limits of Scientific Explana-
1212 tion. *arXiv preprint arXiv:2606.00226*.

1213 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1214 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1215 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

1216 Landauer, R. (1961). Irreversibility and heat generation in the computing
1217 process. *IBM Journal of Research and Development*, 5(3), 183-191.

1218 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1219 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

1220 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1221 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

1222 Barbălat, I. (1959). Systèmes d'équations différentielles d'oscillations non
1223 linéaires. *Revue Roumaine de Mathématiques Pures et Appliquées*, 4, 267-
1224 270.

1225 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1226 of prediction. *Physical Review Letters*, 109(12), 120604.

- 1227 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1228 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- 1229 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1230 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1231 181.
- 1232 Coddington, E. A., & Levinson, N. (1955). Theory of Ordinary Differential
1233 Equations. *McGraw-Hill*, New York.
- 1234 Cover, T. M., & Thomas, J. A. (2006). Elements of Information Theory (2nd
1235 ed.). *Wiley-Interscience*, Hoboken, NJ.
- 1236 Shenker, O. R. (2000). Logic and entropy. *Philosophy of Science*, 67, S146-
1237 S152.
- 1238 Lacker, D. (2016). Mean field games and interacting particle systems.
1239 *Preprint*.
- 1240 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *Lecture Notes in*
1241 *Mathematics*, vol. 1464. Springer.
- 1242 Edelman, G. M., & Tononi, G. (2000). *A Universe of Consciousness: How*
1243 *Matter Becomes Imagination*. Basic Books.
- 1244 Tononi, G., & Edelman, G. M. (1998). Consciousness and complexity. *Sci-*
1245 *ence*, 282(5395), 1846–1851.

- 1246 Friston, K. (2005). A theory of cortical responses. *Philosophical Transactions*
1247 *of the Royal Society B*, 360(1456), 815–836.
- 1248 Strogatz, S. H., & Mirollo, R. E. (1991). Stability of incoherence in a popula-
1249 tion of coupled oscillators. *Journal of Statistical Physics*, 63(3-4), 613–635.
- 1250 Buzsáki, G., & Draguhn, A. (2004). Neuronal oscillations in cortical net-
1251 works. *Science*, 304(5679), 1926-1929.
- 1252 Lindén, H., Pettersen, K. H., & Einevoll, G. T. (2011). Modeling the spatial
1253 reach of the local field potential. *Neuron*, 72(5), 859-872.

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